# Slope Fields & Euler's Method

MATH 146 – Calculus II

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# Differential Equations: A Graphical Perspective

Given a first order equation of the form $y' = f(x, y)$, we may not be able to solve it, but we know the slope of the tangent line to the solution at *every* point in the domain.

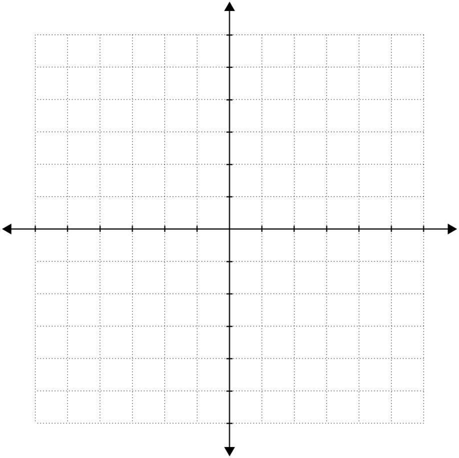
**Example:** Find the slope of the solution to the differential equation $y' + y = x$ at the point $(0, 1)$.

# Slope Fields

## Slope Fields

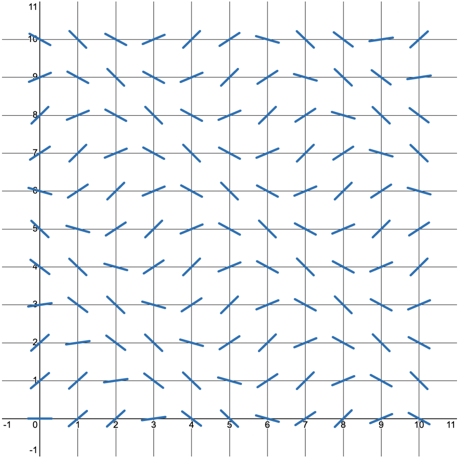
A **slope field** for a differential equation $y' = f(x, y)$ is a graphical representation of the differential equation consisting of a vector field at a (uniformly-spaced) array of points $(x, y)$, where all vectors have the same length and the slope of the vector at $(x_0, y_0)$ is
$y'(x_0, y_0) = f(x_0, y_0)$.

**Example:** Plot a slope field for $y' = x - y$.



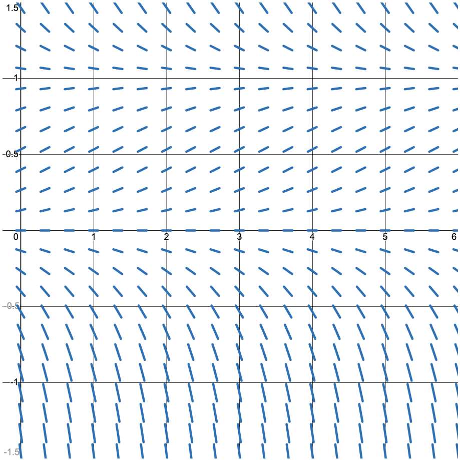
# Slope Fields

**Example:** Plot a slope field for $y' = \sin(x + y)$, $y(0) = 3$.



# Slope Fields

**Example:** Plot a slope field for $y' = y(1-y)$.



# Recall: Linear Approximation

## Linear Approximation

If $f(x)$ is a function, then the **linear approximation** to $f(x)$ at $x = a$ is the line $L(x) = f(a) + f'(a)(x - a)$.

Recall:

- This is just the tangent line to $f(x)$ at $x = a$.
- The tangent line to $f(x)$ at $x = a$ gives a pretty good approximation to $f(x)$ nearby to $a$.

**Example:** Find the linear approximation to $f(x) = x^2$ at $a = 2$.

# Linear Approximation and Differential Equations

Given an initial value problem $y' = f(x, y)$, $y(x_0) = y_0$, you have everything you need to find the linear approximation to the solution $y(x)$ at the point $x_0$.

**Example:** Find the linear approximation to $y' = x - y$, $y(0) = 1$ at $a = 0$.

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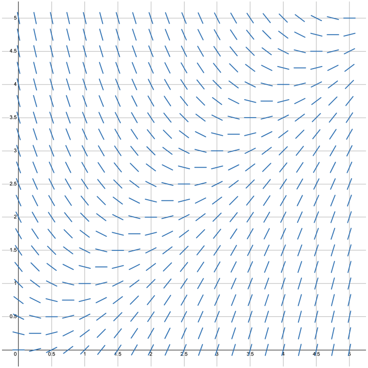


Get off that linear approximation and onto a new one at some point $x_0 + h$. Rinse and repeat.

# Euler's Method

## Euler's Method

To estimate the solution $y(x)$ to the differential equation $y' = f(x, y)$ with the initial condition $y(x_0) = y_0$ at the point $x_n = x_0 + hn$, where $h$ is the "step size" of the method, proceed step-by-step:

$$
\begin{array}{ll}
x_0 \text{ (given)} & y_0 = y(x_0) \text{ (given)} \\
x_1 = x_0 + h & y_1 = y_0 + f(x_0,\, y_0)h \\
x_2 = x_1 + h & y_2 = y_1 + f(x_1,\, y_1)h \\
x_3 = x_2 + h & y_3 = y_2 + f(x_2,\, y_2)h \\
\vdots & \vdots \\
x_n = x_{n-1} + h & y_n = y_{n-1} + f(x_{n-1},\, y_{n-1})h
\end{array}
$$

# Slope Fields

**Example:** Find an estimate for $y(1)$ for the initial value problem $y(0) = 1$. Use step size $h = 0.5$.